

# CS-768 (Learning with Graphs) Course Project Report

## GraphTransLWG: Hybrid GNN–Transformer for Long-Range Graph Reasoning

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## 1 Introduction

Graph Neural Networks (GNNs) aggregate neighborhood information using message passing, but depth increases can cause oversmoothing and oversquashing. This limits long-range dependency modeling. Our project studies a GraphTrans-style hybrid architecture that interleaves local graph convolutions with global self-attention to capture both local topology and distant interactions.

## 2 Problem Setup

Given a graph  $G = (V, E)$  with node features  $X \in \mathbb{R}^{|V| \times d_x}$  and graph label  $y$ , learn  $f_\theta(G) \rightarrow \hat{y}$  for graph-level prediction.

**Message Passing (local bias).** A generic layer updates node  $i$  as:

$$\mathbf{h}_i^{(\ell+1)} = \phi\left(\mathbf{h}_i^{(\ell)}, \square_{j \in \mathcal{N}(i)} \psi(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij})\right),$$

where  $\square$  is an aggregation operator.

**Global attention (long-range context).** With hidden matrix  $H \in \mathbb{R}^{n \times d}$ :

$$Q = HW_Q, \quad K = HW_K, \quad V = HW_V,$$

$$\text{Attn}(H) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right)V.$$

This enables every node to interact with every other node in one layer.

## 3 Architecture

The implemented model follows:

1. Input projection:  $X \mapsto H^{(0)} \in \mathbb{R}^{n \times d}$ .
2. Add a graph-level [CLS] token per graph for pooling.
3. Repeat for  $L$  blocks:
  - GNN sublayer for local structure encoding.
  - Multi-head self-attention sublayer for global context.
  - Position-wise MLP sublayer.
  - Residual paths and normalization in transformer stack.
4. Readout from [CLS] embedding and linear head for logits.

### 3.1 The input projection layer

This layer involves a linear transformation of the input node features to the embedding dimension  $E$ , typically of a higher dimension. This does not account for the discreteness of values of individual input features (that will explode the input dimension). Further, CLS tokens are initialized as a learnable parameter vector of same embedding dimension  $E$ , and have no input features. For a batch, one CLS token is added per graph.

### 3.2 The GNN sublayer

Each transformer layer contains a GNN, an Attention Layer and an MLP. The GNN layer typically has 2 sub-layers of message passing. The CLS tokens are completely ignored in the GNN layer and the original embedding is passed directly to the attention layer.

Each GNN layer does the following math, with  $\vec{h}_{l,u}$  representing the embedding of node  $u$  after layer  $l$  (where  $l = 0$  signifies the input embeddings per node and  $\mathcal{N}(u)$  is the set of neighbors of node  $u$ ):

$$\vec{h}_{l,u} = \vec{h}_{l,u} @ \mathbf{W}_{1,l}^T \quad (1)$$

$$\vec{e}_{l;(u,v)} = (\vec{h}_{l,u} \| \vec{h}_{l,v}) @ \mathbf{W}_{2,l}^T \quad (2)$$

$$\vec{n}_{l,u} = \sum_{v \in \mathcal{N}(u)} \vec{e}_{l;(u,v)} \quad (3)$$

$$\vec{m}_{l,u} = \sum_{v \in \mathcal{N}(u)} \vec{h}_{l,v} \quad (4)$$

$$\vec{g}_{l,u} = \text{GELU}(\vec{h}_{l,u} \| \vec{n}_{l,u} \| \vec{m}_{l,u}) \quad (5)$$

$$\vec{h}_{l+1,u} = \vec{g}_{l,u} @ \mathbf{W}_{4,l}^T \quad (6)$$

### 3.3 The attention sublayer

The attention sublayer is a standard multi-head self-attention mechanism. The CLS token is included in the attention computation, allowing it to attend to all nodes and aggregate global information. The attention output is then passed through a position-wise MLP and residual connections before the next block.

Note that among other non-CLS nodes, the attention mask may be restricted or scaled to reduce importance of interaction among distant nodes, but the CLS token can attend to all nodes without restriction.

Also, cross-graph attention in a batch is disabled.

### 3.4 The MLP sublayer

The MLP sublayer consists of a  $L_{MLP} = 2$ -layer feedforward network with a non-linear activation (e.g., GELU) in between. It is applied position-wise to the output of the attention sublayer, meaning it processes each node's embedding independently.

Separate set of parameters are used for the CLS token for correctly performing computations on the CLS token embedding (may be different from the computations on node embeddings).

## 4 Implementation Mapping to Repository

- `models/full_model.py`: `GraphTransConfig`, `GraphTransModel`, `ModelTrainConfig`, CLS pooling path.
- `models/transformer.py`: transformer block composition.

- `models/gnn.py`: local message-passing layers.
- `models/attention.py`: attention mechanism implementation.
- `models/train.py`: training loop with optimizer/loss/backprop.
- `data_utils/config_objects.py`: dataset and training configuration objects.
- `data_utils/dataset_utils.py`: dataset loading and preprocessing utilities.
- `trainers/base_trainer.py`: generic trainer with grad clipping + scheduler hooks.
- `trainers/flag_trainer.py`: FLAG adversarial perturbation training.

## 5 Training Objective and Optimization

For classification, with logits  $z_g = f_\theta(G_g)$  and labels  $y_g$ , we minimize:

$$\mathcal{L}(\theta) = \frac{1}{B} \sum_{g=1}^B \text{CE}(z_g, y_g).$$

Training uses Adam-family optimization with optional gradient clipping:

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}.$$

### 5.1 FLAG Robust Training

FLAG injects adversarial feature perturbations  $\delta$  in hidden/node input space:

$$\delta^{t+1} = \Pi_\epsilon \left( \delta^t + \alpha \text{sign}(\nabla_\delta \mathcal{L}(\theta, \delta^t)) \right).$$

Model parameters are updated against perturbed examples, improving robustness and regularization.

## 6 Complexity Discussion

Let  $n = |V|$ , hidden width  $d$ , edges  $m = |E|$ .

- GNN layer cost is typically  $\mathcal{O}(md)$ .
- Full attention cost is  $\mathcal{O}(n^2d)$  in time and  $\mathcal{O}(n^2)$  in memory for attention scores.

Hence, hybrid architectures trade better long-range modeling for higher cost on large graphs.

## 7 Benchmarks and Experimental Scope

Configuration files indicate evaluation on OGBG-Code2, OGBG-MolPCBA, NCI1, and NCI109 with multiple variants (GCN/GIN/PNA with and without virtual node settings).

## 8 Conclusion

This project establishes a technical and reproducible foundation for graph representation learning with GraphTrans-style hybrids. The key benefit is combining local inductive bias from message passing with global receptive fields from attention.